Compositional Perturbation Planning with a Generative Structural Causal World Model

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Abstract

Multi-gene intervention planning is critical to achieve a desired final cell state from either an embryonic stem or an intermediary cell state, while modeling epistatic gene interactions. Current perturbation foundation models are not reliably trained to predict multi-gene perturbations and are conditional-mean estimators rather than intervention operators, so composing perturbations under a reinforcement learning (RL) policy queries the model off-distribution. Here we propose a generative structural causal world model in which transcription factor (TF) perturbations are modeled as latent interventions with support for multi-perturbations where any k-factor combination is solved in closed form rather than rolled out, making model exact and compositional under unseen combinations. On held-out gene knockouts, the model beats baselines on both single and combination perturbations with up to eight TF cocktails, and it predicts held-out pairwise epistasis under leave-one-TF-out cross-validation. This non-additive interaction is learned from incorporating combination-TFs in training through a unique interaction head routed through the latent graph, leading to the model scoring held-out experimentally validated epistatic TF combinations high for their predicted interaction. Given this trained causal world model identifiable under novel multi-gene perturbations, we built a planner agent to learn transformations from human embryonic stem cells (hESCs) to downstream annotated fates including hematopoietic stem and progenitor cells (HSPC). We implement a goal-conditioned GFlowNet policy that proposes TF cocktails scored by the frozen world model's do-operator. The policy generalizes to held-out fate signatures and recovers held-out driver TFs of the hESC-to-HSPC screen, matching model-predictive-control (MPC) search at a fraction of the cost while proposing far more diverse cocktails, and it surfaces canonical hematopoietic master regulators including TAL1, KLF2, and HOXB4 on a zero-shot HSPC fate.

1 Introduction

Directing a cell to a target fate is the central problem of cellular reprogramming, and it is combinatorial as fates are reached by overexpressing curated sets of transcription factors (TFs), delivered together at high multiplicity of infection (MOI) and read out days later as a differentiated state [Joung et al., 2023, Briggs et al., 2026]. The cocktail design space is astronomically large, making cell-fate design a natural target for a model-based planner, proposing TF cocktails for unseen fates. Such a planner needs a world model that predicts the outcome of novel and simultaneous multi-TF interventions. Current perturbation foundation models fit this role poorly. scGPT, scFoundation, GEARS, and STATE [Cui et al., 2024, Hao et al., 2024, Roohani et al., 2024, Adduri et al., 2025] are trained as conditional-mean predictors that transfer a perturbation's effect across cellular contexts. In this high MOI combinatorial regime, these models do not reliably beat additive or linear

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baselines [Ahlmann-Eltze et al., 2025, Wu et al., 2024]. This causes a more critical issue for planner or reinforcement learning agents, as these models, acting as amortized conditionals rather than intervention operators, give no compositional or identifiability guarantee for unseen combinations, so a policy that composes TFs veers the model off-distribution and compounds single-step error. Trained for single-step conditional means, they also do not capture the simultaneous high-MOI biology of these screens.

Similar work, like DiRL, couples goal-conditioned RL with a single-cell foundation model as a differentiable world model [Chang et al., 2025]. There, however, the world model, the reward, and the success metric all live in one foundation-model embedding space with no external ground truth, and the reward is proximity in the model's own latents, validated only post hoc, which forces the foundation model itself to stand in as ground truth, a strong and untested assumption. Here we instead require a world model whose structure is testable, exact and identifiable under multiple perturbations, and a planner scored against experimental driver answer keys. We use two human embryonic stem cell (hESC) screens that separate complementary signals. Joung et al. overexpresses single TFs, read out at an early differentiating intermediate (day 7), isolating each TF's directional effect [Joung et al., 2023], and Briggs et al. overexpresses TF *combinations* at high MOI, read out at a committed hematopoietic stem and progenitor cells (HSPC)-like state (days 14–28), giving true combinatorial fate outcomes [Briggs et al., 2026]. Target-fate annotations, apart from the recovered HSPC fate, come from the human fetal cell atlas for both Joung and Briggs [Cao et al., 2020].

Contributions. (1) We build a generative structural-causal world model that predicts the effect of a k-TF intervention in one closed-form solve (Eq. 2). The model composes any unseen combination directly from its single and combo-TF training data, predicting perturbation outcomes of novel combinations. The model beats a fair additive baseline on held-out single and combination perturbations and predicts held-out epistasis on experimentally validated TF pairs. **(2)** We pair this world model with a goal-conditioned GFlowNet planner that samples diverse TF cocktails scored by the frozen do-operator, recovering key TF drivers. **(3)** Lastly, we evaluate the planner against experimental driver lists, finding that it matches model-predictive control (MPC) search at far lower cost with greater cocktail diversity, and surfaces held-out and zero-shot hematopoietic drivers.

2 A structural causal world model

Latent structural equation model. We represent a cell as a latent state $z \in \mathbb{R}^d$ over d=357 gene programs, one per Gene Ontology biological-process (GO:BP) gene set. An encoder $q_\phi(z \mid x)$ maps a cell's 4366-gene expression x to z. The programs follow a linear structural equation model (SEM) on a learned weighted directed graph $A \in \mathbb{R}^{d\times d}$ driven by an exogenous term ϵ,

$$z = A^\top z + \epsilon, \qquad z_0 = (I - A^\top)^{-1}\epsilon_0 \tag{1}$$

A linear decoder reads programs back to genes, $\hat{x} = Dz$ with $D \in \mathbb{R}^{4366\times d}$. Following the masked-decoder (SENA) construction [de la Fuente et al., 2025, Zhang et al., 2023], column j of D is tied to program j by a GO:BP membership mask M. The effective weight is $D \odot M$ (i.e. a gated decoder), so a gene loads fully on programs that have that gene enriched for and shrunk by a leak factor λ=0.3 for unannotated programs. This pins each latent axis to a named program and forces gene covariance to route through the programs and A rather than through free gene-gene weights.

Perturbation as a soft intervention. Here we represent the overexpressing transcription factor (TF) g enters as a program-space shift, or shift encoder [put LCCVAE citation here], $S_g \in \mathbb{R}^d$ added to the exogenous term, $S_g = f_\theta(e_g)$. The descriptor e_g is a 100-dimensional embedding of the TF's amino-acid sequence from the PCA-reduced and z-scored embedding of that gene from the ESM2 protein language model [Lin et al., 2023]. ESM2 embeds any TF, including held-out ones, which is what lets a shift transfer zero-shot to unseen factors, and sequence-similar TFs receive similar shifts. To predict the latent effect of this intervention shift, we introduce a per-program mutilation gate $u \in [0,1]^d$ on the latent graph *A*, that allows for a learned continuous edge weight scaling of the latent edge as opposed to full removal, through scaling the incoming edges of an intervened program by $(1-u)$.

Exact, compositional multi-perturbation. For a set $\mathcal{P}$ of perturbed TFs the intervened latent z_{post} is derived from the following closed-form solve,

$$z_{\text{post}} = \big(I - \operatorname{diag}(1-u)\,A^\top\big)^{-1}\Big(\epsilon_0 + \textstyle\sum_{g\in\mathcal{P}} S_g + S_{\text{int}}(\mathcal{P})\Big), \qquad \hat{x} = Dz_{\text{post}}, \tag{2}$$

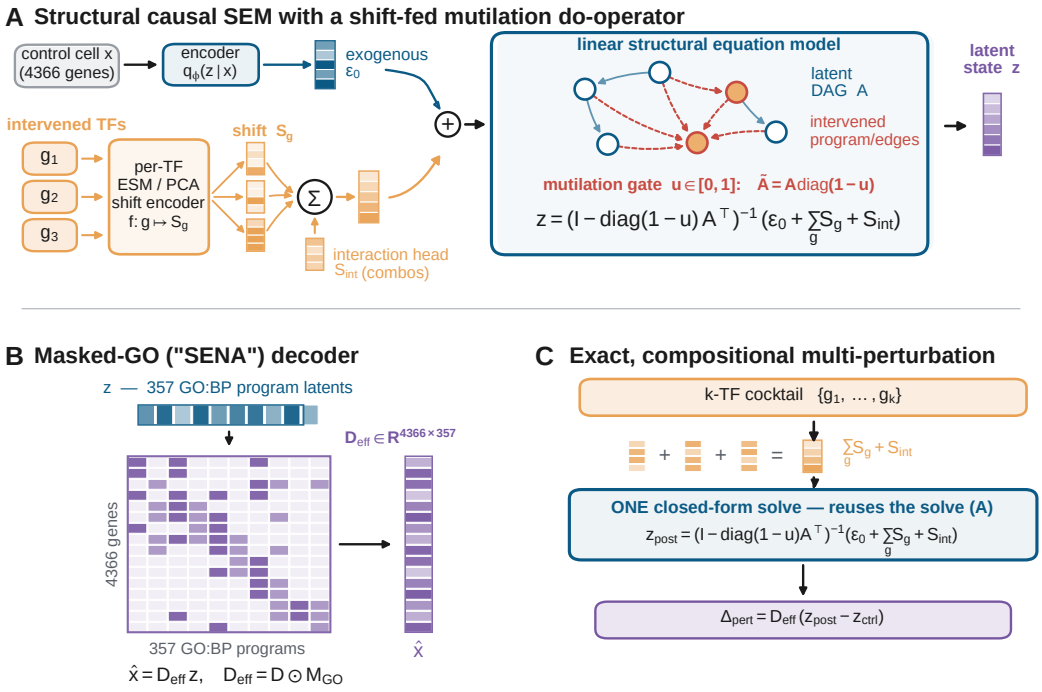


Figure 1: **Structural causal world model.** (A) Latent-DAG SEM with the shift-mutilate do-operator. (B) Masked gene-program (SENA) decoder $x{=}Dz$, columns tied to GO:BP programs. (C) A k-TF perturbation is one closed-form solve of the summed per-TF shifts, so prediction is exact, compositional, and zero-shot.

with $u = \max_{g \in \mathcal{P}} u_g$. A k-TF prediction is the exact solution of the same SEM, built from the same per-TF shifts used for singles, so it is compositional and zero-shot with no combination-specific parameters. We also introduced an interaction term $S_{\text{int}}(\mathcal{P})$ which is a low-rank ($r{=}32$) interaction head that adds a set-dependent latent shift to the total interventional shift. It is zero-initialised and trained only on combinations ($|\mathcal{P}| \geq 2$), so it vanishes for single TFs and supplies the non-additive part of a combination, allowing us to model epistatic interaction routed through the latent graph A.

Training objective We train the model once for 60 epochs with

$$\mathcal{L} = \mathcal{L}_{\text{rec}} + \beta\, \mathcal{L}_{\text{KL}} + \lambda_{\text{pert}}\, \mathcal{L}_{\text{pert}} + \lambda_{\text{spec}}\, \mathcal{L}_{\text{spec}}. \tag{3}$$

$\mathcal{L}_{\text{rec}}{=}\|\hat{x} - x\|^2$ is the reconstruction MSE and $\mathcal{L}_{\text{KL}}$ the diagonal-Gaussian VAE term with β annealed $0.01{\to}0.5$. The perturbation term ($\lambda_{\text{pert}}{=}10$) is computed on cell sentences of 64 cells so its target is a per-perturbation pseudobulk change Δ, and it sums four parts. A mean-shift MSE $\|\Delta_{\text{pred}} - \Delta_{\text{obs}}\|^2$ matches the average response. A Gaussian-RBF maximum-mean-discrepancy MMD^2 matches the full predicted and observed perturbed-cell distributions, not only their mean. A symmetric InfoNCE term (temperature 0.2) pushes each predicted profile toward its own perturbation and away from others. A DE-gene MSE weighted $8\times$ preserves per-gene response magnitude on the differentially expressed genes. The spectral term $\mathcal{L}_{\text{spec}}{=}\text{relu}(\sigma_{\max}(A) - 0.95)$ keeps the SEM solve well-conditioned. More details on the architecture can be found in Appendix A.

3 Prediction, non-additivity, and a load-bearing causal graph

Setup and training. We train the world model on two overexpression screens, the Joung single-TF atlas and the Briggs combinatorial TF screen, over a shared 4366-gene axis with $d{=}357$ GO:BP program latents, trained in two stages. First we fit a base model on single-TF perturbations, which learns one shift per TF. Then we finetune this base on TF combinations, and we oversample the single-TF examples $8\times$ so the model keeps its single-TF accuracy as it learns combinations. We

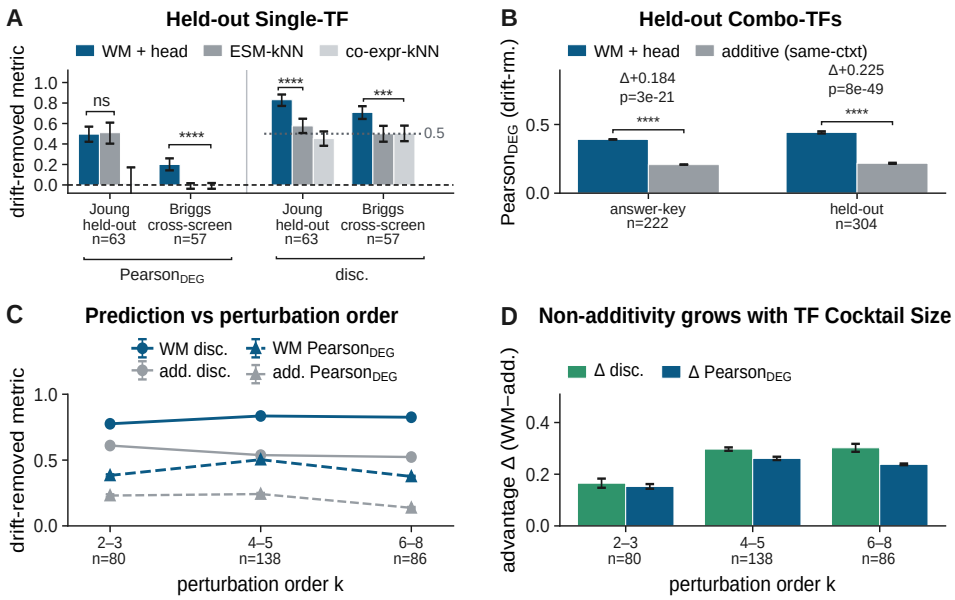


Figure 2: **Held-out prediction beats a fair additive baseline.** (A) Single-TF prediction against ESM- and co-expression-retrieval baselines (drift-removed $\mathrm{Pearson}_{\mathrm{DEG}}$and discrimination). (B) Held-out combinations against the same-context additive baseline. (C) Metrics against perturbation order k. (D) The world-model advantage grows with order. Mean over three seeds, SEM error bars, n per panel.

build two checkpoints from this recipe. The deployable checkpoint trains on all 495 single-TF perturbations and all 1622 combinations (arrayed Joung pairs and pooled Briggs sets of up to eight TFs) and holds out only the 222 Briggs HSPC-reaching validated combinations. However, for the held out predictions shown in Fig. 2, we train a similar model with held out single and combination TFs with a $85\%/80\%$ single/combination split across three seeds, alongside sampled replays of earlier training data during finetuning to ensure model won't forget single-TF level prediction, and we report the held out values on this sibling model. For the downstream policy training, we split fate targets by fate name into disjoint folds and always hold out the HSPC fate. Appendix B gives the full data, splits, and hyperparameters along with methodological details.

Evaluating the World Model Overexpressing any TF at high MOI produces a large response that is largely shared across factors a generic overexpression signature, pronounced here because many assayed TFs are master regulators. This shared component is a confound: a model that predicts it for every perturbation scores well on raw correlation while distinguishing no TF from any other, which is precisely the mechanism by which deep models tie simple baselines on published benchmarks [Ahlmann-Eltze et al., 2025]. We therefore evaluate on the TF-specific residual. We estimate the drift as the mean single-TF Δ over the training set only, subtract it from both prediction and observation, and report two drift-removed quantities. $\mathrm{Pearson}_{\mathrm{DEG}}$ is the Pearson correlation between the drift-removed predicted and observed change on the perturbation's differentially expressed (DE) genes (log-fold-change above 0.1, with the perturbed TF excluded), and measures whether the predicted response has the right shape. Discrimination (disc) is the area under the ROC of matching a prediction to its own held-out perturbation against all others by drift-removed similarity, and measures whether the response is specific to the right TF rather than a generic perturbation signature. We compare against a same-context additive baseline that sums observed single-TF Δ vectors, a no-change baseline, and a context-mean baseline [Ahlmann-Eltze et al., 2025]. For single-TF prediction we add two retrieval baselines that copy the drift-removed response of the nearest training TF, by ESM2 similarity or by co-expression.

Held-out single-TF prediction. On held-out single TFs the world model is competitive with the strongest baseline and far more specific (Fig. 2A). On held-out Joung single TFs it matches the ESM2 nearest-neighbour copy in $\mathrm{Pearson}_{\mathrm{DEG}}$ (0.50 against 0.51) but separates the perturbations

much better in discrimination (0.83 against 0.58), and on Briggs single-TF effects, a cross-screen test to see whether model, that only saw Briggs combinations, can still generalize to singles, it outperforms every baseline on both metrics for single TF perturbation prediction.

The world model beats a fair additive baseline on combinations. Now, testing the models performance on predicting combos of TFs given a cocktail set S, on held-out combinations, the model beats the same-context additive baseline by 0.18 to 0.23 in $\mathrm{Pearson}_{\mathrm{DEG}}$, a gain of $+0.184$ on the 222 held-out answer-key combinations (p=2.8×10^{-21}) and $+0.225$ on the 304 held-out combinations (p=8.4×10^{-49}) over three seeds (Fig. 2B). The advantage persists to high order. At k=6 to 8 TFs the model reaches discrimination 0.84 and $\mathrm{Pearson}_{\mathrm{DEG}}$ 0.38 against 0.53 and 0.15 for the additive baseline (n=86, p=3.1×10^{-15}), and it beats additive on 95% of held-out combinations (Fig. 2C,D), so the non-additive margin grows with combination order.

Epistasis is captured, and it comes from the interaction head. We measure the interaction, or net epistasis, as the combination response minus the sum of its single-TF responses. On held-out combinations the model predicts the observed interaction at Pearson r=0.70 over n=276 Briggs combinations (Fig. 3A). The signal is concentrated in higher order. Broken out by order, the held-out concordance is near chance for pairs and triples (k=2–3, r=0.03) and strong for larger cocktails, reaching r=0.66 at k=4–5 and r=0.83 at k=6–8 (Fig. 3C). This is exactly the part the model learns through the rank-32 interaction head of §2. A singles-only model, whose head is never trained and stays zero, captures none of it (r_{pair}=0.06), while the combination-trained model reaches r_{pair}=0.70 (Fig. 3B). The model recovers the sign and ranking of the interaction but not its magnitude, which shows up as a sub-additive offset we report rather than hide.

Is the causal graph load-bearing? Prediction accuracy alone does not tell us whether the latent structural causal graph A is actually *used* for prediction, or whether it is an inert byproduct of training. We test this by ablating A inside the frozen do-operator and re-measuring prediction, and we then characterize the biological structure of the learned graph.

The directed graph is load-bearing for prediction (Fig. 3E). We ablate A inside the frozen do-operator by reversing, symmetrizing, or shuffling its edges, and re-measure Δ-prediction on the deployable model (a relative true-vs-shuffled comparison on the same perturbations, so it does not need a held-out split). Against a 1000-shuffle null, single-TF prediction collapses from Pearson $0.536\,[0.443, 0.625]$ to a shuffled mean of -0.156 ± 0.010, a gap of 0.692 that no shuffle approaches (permutation $p < 0.001$, $0/1000$, paired Wilcoxon across 66 TFs p=8.9×10^{-11}, App. H), and 59 of 66 TFs individually predict worse under a shuffled graph (App. H). Reversing the direction of edges is as damaging as shuffling them, so the dependence is on edge *direction*, not connectivity alone, the signature of the continuous do-operation and interventional training. The same directional dependence holds on combinations, where all 14 held-out sets predict worse under a shuffled graph (gap 0.102, permutation $p < 0.001$, Wilcoxon p=6.1×10^{-5}). The effect is consistent across metrics as well. Reversing or shuffling A collapses cross-fate discrimination AUROC from 0.84 to 0.60 for single TFs (Fig. 3E,F), and top-1 driver recall from 0.73 to 0.02 (App. H).

The latent graph is biologically structured. The masked decoder pins each latent axis to a named GO:BP program at mask coherence 1.00, and the learned adjacency A is a sparse signed graph over those programs whose highest-connectivity hubs are developmental programs. Compared against known biological priors (CollecTRI, DoRothEA, STRING), the projected graph carries above-chance signal but does not beat a plain co-expression control, as expected for a context-specific model trained prior-off on TFs alone. We therefore make no claim of literal GRN recovery and restrict the causal claim to prediction (Sec. 7).

4 A goal-conditioned GFlowNet planner

Planning against the frozen causal world model. Given a target fate we want to propose TF cocktails that steer a control cell toward it, without re-running a search for every new fate. We train a goal-conditioned policy that proposes a cocktail. The frozen do-operator of §2 predicts the resulting Δ state by propagating the injected shifts over the learned causal graph A, and the reward $\Phi(x, g)$ is the drift-removed agreement of that predicted Δ with the target fate's observed signature (Fig. 4). The causal graph is the planner's prior: through the do-operator it defines which cocktails are predicted to reach which fates. At deployment the policy proposes candidates directly, with no per-fate search.

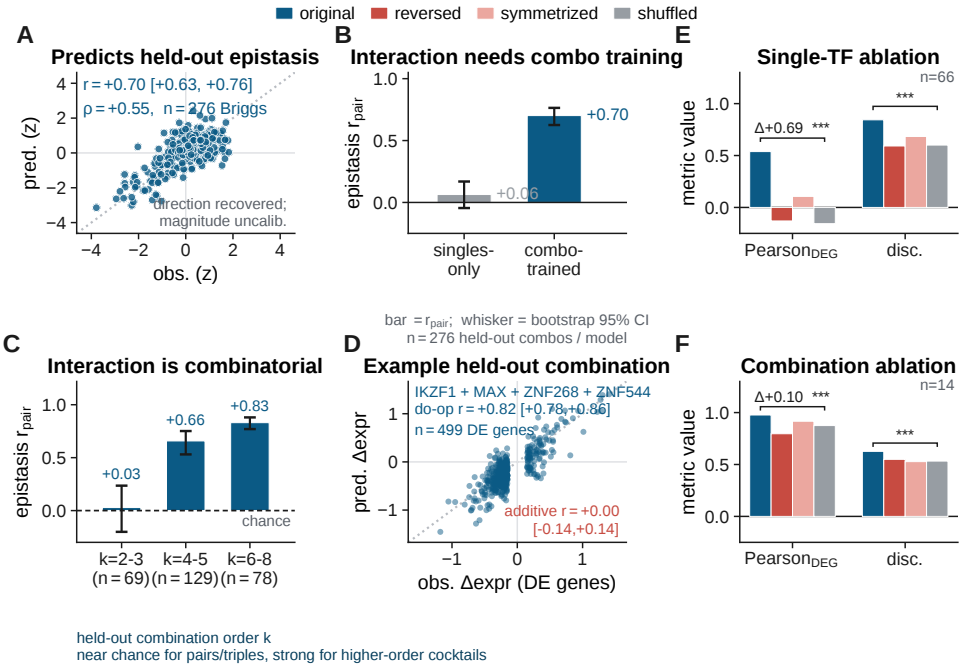


Figure 3: **Combinatorial epistasis, and a load-bearing causal graph.** (A) Predicted against observed interaction on n=276 held-out Briggs combinations (r=0.70). (B) A singles-only model captures none. (C) Concordance rises with combination order. (D) An example held-out combination. (E, F) Single-TF and combination prediction under original, reversed, symmetrized, and shuffled A, over n=66 held-out single transcription factors and n=14 held-out combinations. Bars are point estimates tested against a 1000-shuffle-A permutation null (stars) and a paired Wilcoxon over the held-out units (p=9×10^{-11} and 6×10^{-5}). Reversing the edges is as damaging as shuffling, so the dependence is directional (drift-removed Pearson gaps of 0.69 and 0.10 over the null).

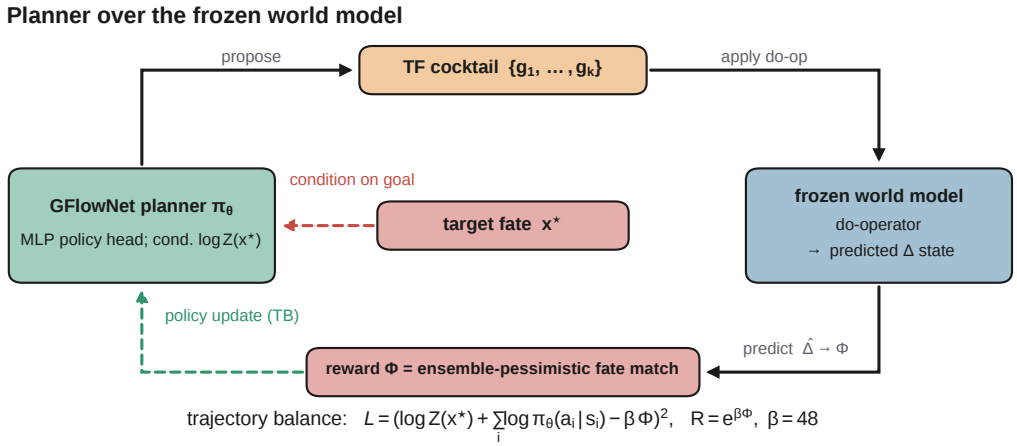


Figure 4: **Goal-conditioned GFlowNet planner.** A target fate conditions a trajectory-balance GFlowNet that proposes a TF cocktail. The frozen do-operator propagates the injected shifts over the learned causal graph A to predict the Δ state, and the reward measures closeness to the target. The policy is trained by the trajectory-balance gradient with a goal-conditioned $\log Z$ head.

Why a GFlowNet, conditioned on the goal. Many distinct TF sets can steer a cell to the same fate, so we want a planner that *samples* diverse cocktails in proportion to reward rather than collapsing

onto a single mode. We train a trajectory-balance GFlowNet [Bengio et al., 2021] whose partition-function head is conditioned on the target fate, $\log Z(g)$. This goal-conditioning is what lets the policy spread its mass across modes. A shared $\log Z$ head instead collapses onto the reward argmax (App. E).

An ensemble-pessimistic reward. Because the world model is an imperfect oracle that a planner could exploit, we score each cocktail under an ensemble of $K{=}6$ independently trained world models and reward the pessimistic estimate $\Phi_{\text{pess}} = \text{mean}_k\, \Phi^{(k)} - \lambda\, \text{std}_k\, \Phi^{(k)}$, which penalizes cocktails the ensemble disagrees on. A modest $\lambda{=}0.25$ gave the best recovery–diversity trade-off. The policy head is a plain per-slot multilayer perceptron over the shared candidate vocabulary. Full training and hyperparameter details are in App. C and the complete cross-arm evaluation in App. E.

5 Cell-fate design results

Protocol. We evaluate the planner on four target-fate sources (Joung single- and combination-defined fates, the Briggs hESC-to-HSPC screen, and a held-out zero-shot HSPC fate) under a leakage-clean fold-by-name split over five seeds and three folds. The policy draws $N{=}256$ candidate cocktails per fate, read out by their elite (top-decile by reward) frequency at no extra world-model cost. We compare against budget-matched baselines: model-predictive control (MPC, beam search over the same shortlist, $\approx$9,240 world-model evaluations per fate), a greedy beam-1 variant, an additive single-TF baseline, and random shortlists. Because the elite readout is reward-guided, its chance level is a reward-matched null, and we test each method's gain over it with a paired Wilcoxon across held-out fates. App. D defines every metric.

Recovery matches exact search and dominates the heuristics, at a fraction of the compute (Fig. 5A,B). On the 41 held-out Briggs fates the policy beats the budget-matched additive and random baselines on every metric by a wide margin (paired per-fate Wilcoxon $p < 10^{-4}$ throughout), recovering 0.58 of the answer-key drivers at elite recall@20 against 0.01 and 0.07 for the additive and random baselines. Against exact model-predictive control (MPC) beam search it is at parity on the broad metrics, elite recall@20 0.58 against 0.51 and soft driver recall@20 0.80 against 0.72 (paired Wilcoxon $p{=}0.1$ and 0.2, not significant, Fig. 5B), and it is significantly *ahead* at the top of the ranking, recall@5 0.36 against 0.24 and precision@5 0.20 against 0.13 (both $p < 0.05$). It reaches this while drawing its 256 cocktails with $\approx$36$\times$ fewer world-model evaluations than beam search, an amortized single pass. By source the recovery concentrates on Briggs, the two Joung sources sitting near the matched-null floor (App. E).

Composition, not addition, is what the planner exploits. The recovered cocktails are large, six to eight transcription factors on the Briggs fates, and their predicted effect is strongly non-additive. This is the same higher-order interaction the world model captures on held-out combinations, where concordance rises from near chance for pairs to $r{=}0.83$ at order six to eight (Fig. 3C), and it is exactly what an additive single-TF baseline cannot represent, which is why that baseline sits at the null. The planner inherits this combinatorial structure through the do-operator and uses it to assemble sets whose joint predicted effect exceeds the sum of their single-TF parts.

The policy is diverse at fixed quality, and search is not (Fig. 5C). For each held-out HSPC fate we draw 256 candidate cocktails from each method and set an arm-independent quality bar at the 75th percentile of the reward of matched-random cocktails, then count how many *distinct* cocktails each method proposes *above* that bar, so the metric credits diversity only among high-reward designs and not a flood of low-quality ones. The policy returns roughly 160 such distinct high-quality cocktails, a random sampler 64, and beam search 1, as beam is a single-solution optimizer rather than a sampler. Diversity at fixed quality, not raw recovery, is where amortization is uniquely useful, handing an experimentalist many equally good candidates to array rather than one.

Zero-shot HSPC is a diversity-coverage advantage (Fig. 5D,E). We test the frozen policy zero-shot on a held-out hESC-to-HSPC fate against the Briggs arrayed answer key (54 driver TFs across 28 arrayed-validated combinations). On this broad key the policy's diverse elite proposals cover far more of the answer key than search or the heuristic baselines, an arrayed recall@50 of 0.13 against $\leq$ 0.06 and 15 of the 54 drivers surfaced across runs against 3–4 for the baselines (Fig. 5E). The proposed factors are biologically coherent (Fig. 5D): with the regulatory prior switched off the policy nominates the canonical hematopoietic regulators LYL1, KLF1, KLF2, and IKZF1 in nearly every run, and members of the blood stem-cell heptad [Wilson et al., 2010], TAL1/SCL, GATA2,

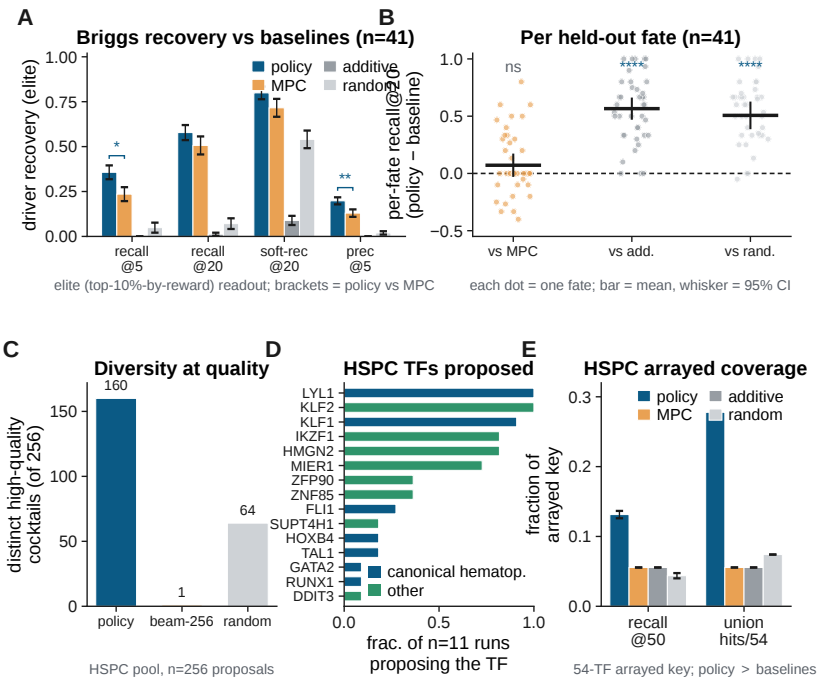


Figure 5: **Cell-fate design: recovery, diversity, and zero-shot HSPC.** Recovery panels use the elite (top-10%-by-reward) readout on n=41 held-out Briggs fates, budget-matched to model-predictive control (MPC), additive, and random baselines on the same fates. (A) Briggs recovery across four metrics (mean±SEM over fates). (B) Per-fate elite recall@20, policy minus each baseline (one dot per fate): parity with MPC (ns), far above additive and random ($p < 10^{-4}$). (C) Distinct high-quality cocktails from 256 HSPC-pool proposals. (D) Arrayed-key TFs the policy proposes zero-shot on the held-out HSPC fate, bar is the fraction of 11 runs proposing each, colored canonical hematopoietic versus other. (E) Coverage of the 54-TF Briggs arrayed key (28 combinations), elite recall@50 and distinct drivers surfaced, policy versus baselines.

RUNX1, and FLI1, together with the stem-cell self-renewal factor HOXB4 [Antonchuk et al., 2002], in a minority of runs. This is a coverage advantage of a diverse proposer over narrow search, not a per-driver precision result. On the narrow twelve-TF named answer key the policy does not beat model-predictive control, and we do not calibrate a breadth-matched null for the arrayed key, both of which we report rather than hide in App. E.

What the design choices buy (App. E). Goal-conditioning the partition function is what spreads sampling across modes, as a shared partition function collapses onto the reward argmax and loses the diversity above. The ensemble-pessimism weight λ trades recovery against diversity, with λ=0.25 at the knee, and the plain per-slot policy head matches heavier alternatives, so the amortization gain is not an artifact of head capacity.

Honest limits. The cocktails point toward the target fate in world-model Δ-direction (own-minus-other cosine $+0.25$ on Briggs, $+0.19$ on HSPC) but do not discriminate fates by answer-key overlap, a cross-fate discrimination wall (policy well below the driver oracle) we report rather than hide. The two Joung sources are not significant, a training-data coverage limit rather than a method failure (Sec. 7).

Takeaway. The planner is an amortized proposer rather than an oracle. On the well-powered Briggs source it reaches the recovery of exact search at a small fraction of the world-model compute and returns two orders of magnitude more distinct high-quality cocktails, and on a fully held-out HSPC fate its diverse proposals cover more of the arrayed answer key with biologically coherent regulators than search does. The two Joung fate sources, with only six and two held-out fates and one-to-five-TF answer keys, are underpowered and sit at the matched null, and where fates share drivers the

policy cannot yet separate them. We read these as evaluation and discrimination frontiers for the next round of screens and models, not as evidence against amortized fate design.

6 Related work

Models of perturbation response. Our world model descends from causal-disentanglement variational autoencoders with soft interventions, including Discrepancy-VAE [Zhang et al., 2023], its gene-ontology-structured variant SENA [de la Fuente et al., 2025], sparse additive mechanism-shift autoencoders [Bereket and Karaletsos, 2023], and Biolord [Piran et al., 2024], all of which amortize the response through an encoder pass. We differ by treating the learned structural equation model as a frozen simulator whose multi-perturbation response is an exact fixed-point solve (Eq. 2), which is what makes zero-shot composition of unseen cocktails well defined. This exact-by-design choice is motivated by audits showing that deep perturbation predictors do not reliably beat simple additive or nearest-neighbor baselines on held-out perturbations [Ahlmann-Eltze et al., 2025, Wu et al., 2024], and training with the regulatory prior off follows benchmarks showing that context-independent priors can hurt context-specific inference [Nourisa et al., 2025].

Planning and amortized diverse sampling. Concurrent work couples goal-conditioned reinforcement learning to a single-cell foundation model [Chang et al., 2025], recovers reprogramming protocols from fitted network dynamics [Wytock and Motter, 2024], or searches directly for reprogramming transcription factors [Li et al., 2025]. These optimize against a learned predictor or search a model once per target. We instead sample cocktails in proportion to a frozen structural-causal reward with a generative flow network [Bengio et al., 2021], whose calibrated diversity fits fate design because many transcription-factor sets reach the same fate. Goal-conditioning the partition function lets one policy serve many fates, turning per-fate search into an amortized proposer.

7 Limitations and conclusion

Limitations. The learned graph is load-bearing for prediction but is not a recovered gene-regulatory network, so we restrict the causal claim to prediction and leave a rotational ambiguity in the adjacency A unresolved (Sec. 3). The captured epistasis is higher-order, near chance for pairs and strong only for larger cocktails, and its magnitude is uncalibrated even where its sign and ranking are right. On planning, the two Joung fate sources are not significant, but that evaluation is underpowered rather than a method failure, with only six single-defined and two combination-defined held-out fates carrying one-to-five-TF answer keys, and the world model is itself trained on the Joung data. A cross-fate discrimination wall also remains for fates defined only by a coarse expression signature. We expect more held-out fates and richer fate representations to resolve both.

Conclusion. A frozen structural causal world model gives exact, compositional, zero-shot multi-perturbation predictions, and a goal-conditioned generative flow network planner over it designs cell-fate cocktails that match exact search at a fraction of the compute, propose far more diverse cocktails, and on a zero-shot HSPC fate cover more of the hematopoietic answer key with biologically coherent regulators than search does. We report where the approach works and where it does not, so the result is usable rather than oversold.

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A Model and objective in full

This section gives the world model and its training objective at the level of exact shapes and hyperparameters. All shapes below are those of the deployed world model at epoch 60. The gene axis has G=4366 genes and the latent has d=357 gene programs. The decoder weight has shape 4366×357, the latent directed graph A has shape 357×357, and the per-TF label matrix has shape 4366×100.

A.1 Encoder and latent variables

The encoder is an amortized Gaussian. It reads a cell expression vector $x \in \mathbb{R}^{4366}$ concatenated with a 16-dimensional modality embedding (Joung is modality 0, Briggs is modality 1) and maps the 4382-dimensional input through a hidden layer of width 512 to a mean μ and log-variance $\log \sigma^2$ over the exogenous term $\epsilon \in \mathbb{R}^{357}$. The first encoder weight therefore has shape 512×4382 and the modality embedding table has shape 2×16. We sample ϵ with the standard reparameterization. The observational forward pass solves the latent structural equation model (SEM) for z and decodes x=Dz+b.

A.2 The latent SEM and its closed-form solve

We model the resting latent state by a linear SEM over a learned signed directed graph $A \in \mathbb{R}^{357\times357}$ driven by ϵ,

$$z \;=\; A^\top z + \epsilon \;\iff\; (I - A^\top)\, z = \epsilon \;\implies\; z = (I - A^\top)^{-1}\epsilon. \tag{4}$$

The closed form follows because $I - A^\top$ is invertible whenever the spectral radius of A is below 1. We enforce that by a power-iteration spectral-norm clamp that rescales every sampled A so $\|A\|_2 \le 0.85$, which guarantees $I - A^\top$ is nonsingular and every downstream solve is well posed. The solve itself is a direct linear solve with a sanitized escalating-ridge fallback for the rare early-training singular case. The adjacency is a factored Gumbel-sigmoid gate times a real-valued weight times a no-self-loop mask, $A_{ij} = \mathrm{gate}_{ij} \cdot w_{ij} \cdot (1 - \delta_{ij})$, with $\mathrm{gate}_{ij} = \sigma((\theta_{ij} + \mathrm{Gumbel})/\tau)$ in training and the hard threshold $\mathbf{1}[\sigma(\theta_{ij}/\tau) > 0.5]$ at evaluation. The temperature τ anneals from 1.0 to 0.25 over training. At evaluation the graph is therefore deterministic and signed.

A.3 The shift-mutilate do-operator and k-TF composition

A perturbation of a transcription factor (TF) g enters as a soft intervention with two parts. First it adds a program-space shift $S_g \in \mathbb{R}^{357}$. Second it applies a per-program mutilation gate $u_g \in [0,1]^{357}$ that partially severs incoming edges. For a set $\mathcal{P}$ of perturbed TFs the deployed operator sums the single-TF shifts, injects the sum into the exogenous term, forms one aggregate gate, and does one mutilated solve. This is Eq. (2) of the main text. Concretely, in the structural intervention mode used throughout,

$$S = \textstyle\sum_{g\in\mathcal{P}} S_g \ (+\, S_{\text{int}}), \qquad \tilde{\epsilon} = \epsilon_0 + S, \tag{5}$$
$$u = \underset{g\in\mathcal{P}}{\text{mean}}\, \text{clip}\big(\beta|S_g|,\, 0,\, 1\big),\ \beta{=}1, \qquad z_{\text{post}} = \big(I - \text{diag}(1-u)A^{\top}\big)^{-1}\tilde{\epsilon}.$$

Two design points matter. The gate u is the per-single mean of $\text{clip}(\beta|S_g|)$ rather than $\text{clip}(\beta|\sum_g S_g|)$. The summed form saturates about 41% of programs to $u{=}1$ on combinations, which detaches A and degrades combination prediction, so we use the per-single mean, which is identical for a single TF. In structural mode the shift is injected only into $\tilde{\epsilon}$, so any non-target program moves only through A. If $A{=}0$ there is no downstream effect, which is what makes A load-bearing for prediction by construction. The operator reduces to the additive shift $z_{\text{prev}} + z_S$ when $u \to 1$ on all driven programs and to predict-control when $u \to 0$. Non-additivity for combinations arises because u, and hence $M = \text{diag}(1{-}u)$, depends nonlinearly on the perturbed set. An optional symmetric low-rank interaction head adds a combination-specific shift $S_{\text{int}} = W_{\text{out}}\big(\frac{1}{2}((\sum_i x_i)^2 - \sum_i x_i^2)\big)$ with $x_i = U(\text{ESM}_i)$ and an interaction rank of 32. S_{int} is 0 for a single gene. The operator is validated against a naive fixed-point solve. A k-TF prediction is therefore one linear solve assembled from the same single-TF shifts used for singles, exact and zero-shot for novel combinations.

A.4 Per-TF shift from ESM embeddings

The shift S_g is produced by a learned shift encoder from a per-gene descriptor ℓ_g. The descriptor is an ESM2 (Evolutionary Scale Modeling v2) protein language model embedding of the TF, reduced by principal component analysis and z-scored to 100 dimensions. The descriptor exists for any TF in the panel, including held-out TFs, which is what lets the shift generalize zero-shot. The encoder is a one-hidden-layer multilayer perceptron (MLP) of width 1024 with a LeakyReLU(0.2) nonlinearity mapping $100 \to 357$, with a raw real-valued output and no output activation. Its first weight has shape 1024×100 and its output weight has shape 357×1024. A second head shares the trunk and emits per-program gate logits, so the shift is $S_g = c \cdot (\text{gate} \odot \text{raw})$ with a learned scalar penetrance c and a sparse target gate that is Gumbel-sigmoid in training and hard at evaluation. The gate open probability is penalized by an L_0 target-sparsity term. A leakage-safe train-only retrieval memory over embedding keys can add an embedding-similarity-weighted shift with the query row self-masked, which we keep off by default in the reported model.

A.5 The masked gene-program (SENA) decoder

The decoder is linear, $x{=}Dz{+}b$ with $D \in \mathbb{R}^{4366\times357}$. Each latent column is tied to a Gene Ontology Biological Process (GO:BP) program by a membership mask $M \in \mathbb{R}^{4366\times357}$. We refer to this masked gene-program decoder scheme as SENA, following [de la Fuente et al., 2025]. The effective decoder weight is the masked-learnable product $D \odot M$, so annotated gene-program pairs learn at full strength while off-annotation pairs are shrunk by a leak factor λ. This forces gene covariance to route through the program latents and A rather than through arbitrary gene-gene edges. The mask is built from the Enrichr GO Biological Process 2023 library. Candidate terms are GO:BP sets with at most 30 genes and at least 5 genes present on our axis. Terms are greedily de-duplicated largest-first so no kept term overlaps a previously kept one by more than 0.5 under the containment overlap $|A \cap B|/\min(|A|,|B|)$. Off-annotation entries take the leak value $\lambda{=}0.3$. The GO:BP selection yields 357 kept programs on our axis, so the latent dimension is set to the number of kept programs and the deployed model has $d{=}357$ program latents. The decoder mask is not frozen, so D is masked-learnable rather than fixed to the mask.

A.6 Training objective, prior-off

The total loss is

$$\mathcal{L} = \lambda_{\text{rec}}\,\mathcal{L}_{\text{rec}} + \beta_{\text{eff}}\,\mathcal{L}_{\text{KL}} + \lambda_{\text{pert}}\,\mathcal{L}_{\text{pert}} + \lambda_{\text{spec}}\,\mathcal{L}_{\text{spec}} + \lambda_{\text{ortho}}\,\mathcal{L}_{\text{ortho}} + \lambda_{\text{sparse}}\,\mathcal{L}_{\text{sparse}} + \lambda_{\text{prior}}\,\mathcal{L}_{\text{prior}}. \tag{6}$$

We train prior-off. The graph prior, decoder-orthogonality, graph-sparsity, decoder-L_1, and acyclicity weights are all zero (λ_{prior}=0, λ_{ortho}=0, λ_{sparse}=0, $\lambda_{\text{D-sparse}}$=0, and both acyclicity coefficients 0), so no external transcription-factor–target database edge is ever used to shape A. The acyclicity term is skipped entirely when its weights are zero to avoid $0 \times$ NaN poisoning. The training log confirms the acyclicity residual stays at 0 throughout. Table 1 lists every active term with its exact form and weight for the deployed model.

The reconstruction term is a plain mean squared error (MSE), modality-gene-masked so a cell only scores genes its modality observes. The Kullback–Leibler (KL) divergence is the standard diagonal-Gaussian variational term with a free-bits floor, and its weight β ramps linearly from 0.01 to 0.5 over the first 30 epochs. The perturbation term $\mathcal{L}_{\text{pert}}$ (weight λ_{pert}=10, clamped at 50 before scaling) is the sum of the sub-terms in Table 1, computed on cell sentences of S=64 cells so the mean-shift target is a per-sentence pseudobulk Δ. The DE-gene MSE sub-term deliberately preserves magnitude and is not whitened, and is restricted to observed differentially expressed (DE) genes above the threshold 0.1, which is the term that carries per-gene response magnitude.

Table 1: Active training-loss terms for the deployed world model. All weights are exact configured values. Terms with weight 0 are omitted (graph prior, decoder-orthogonality, graph-sparsity, decoder-L_1, acyclicity, energy optimal-transport, and shift-L_1). MMD denotes maximum mean discrepancy and InfoNCE the info-noise-contrastive estimation loss.

Term	Form (drift = mean train-single Δ)	Weight	Value
Reconstruction	$\|(\hat{x}-x)\odot m_{\text{mod}}\|^2$ mean-MSE	λ_{rec}	1.0
KL	$-\frac{1}{2}(1+\log\sigma^2-\mu^2-\sigma^2)$, free-bits	β	$0.01 \to 0.5$
Perturbation MSE	$\|\Delta_{\text{pred}}-\Delta_{\text{obs}}\|^2$	λ_{MSE}	1.0
Distribution matching (MMD)	Gaussian-RBF MMD^2, median bandwidth, per sentence	λ_{MMD}	1.0
Goal specificity (InfoNCE)	symmetric NCE on drift-removed norm. Δ, temp 0.2	λ_{NCE}	1.0
DE-gene MSE	unwhitened MSE on $\{\|\Delta_{\text{obs}}-\text{drift}\|>0.1\}$ genes	λ_{DEG}	8.0
Target sparsity (L_0)	$\overline{\sum_k \text{gate-prob}_k}$, ramped	λ_{L_0}	0.03
Spectral	$\text{relu}(\sigma_{\max}(A)-0.95)$	λ_{spec}	0.1

A gradient-reversal dataset classifier on the encoder mean μ drives latent modality-invariance and is added with an implicit coefficient of 1.0 when there is more than one modality. The mutilation is run in shift mode with β=1.0. The mutilation clip of 0.5 is inert in shift mode, where the clamp is to 1.0.

B Datasets, splits, and checkpoint discipline

B.1 Sources

We use three single-cell resources. The Joung TF atlas is a human embryonic stem cell (hESC) single-TF overexpression atlas at day 7 [Joung et al., 2023]. It contributes single-TF perturbations plus a small number of arrayed combinations. The Briggs hESC→fate overexpression screen is a dense combinatorial TF screen read out at days 14 and 28, deposited at GEO under accession GSE328457 [Briggs et al., 2026]. It contributes high-order TF combinations and pluripotent hESC-start controls. The Cao fetal cell atlas defines the target-fate marker signatures [Cao et al., 2020].

B.2 Gene axis and modality alignment

The shared gene axis is the Joung 4366-gene vocabulary. Briggs cells are aligned onto it by matching counts-per-10,000 log-normalized values on shared genes and zero-filling the rest. About 2800 of the 4366 genes are observed in Briggs. A per-modality gene mask keeps the vocabulary at 4366 but scores the reconstruction loss only on genes each modality observes. Joung sees all 4366 genes,

Briggs only its observed subset. Each Briggs cell carries an in-vocabulary TF set obtained by collapsing isoforms and keeping the modelable TFs. Cells are grouped by their in-vocabulary TF set. Sets of in-vocabulary size 2 to 8 with at least 5 replicate cells become combination perturbations. The per-cell median in-vocabulary set size is about 5 and reaches into the teens, so Briggs supplies the high-order regime.

B.3 Answer-key holdout and leakage-free splits

The answer key is held out of training so recovery is honest. A Briggs combination is held out if it shares at least two members with P_1 or P_2, is one of the six canonical hematopoietic stem and progenitor cell (HSPC) pairs, or is one of the arrayed-validated combinations from the supplementary table. The two validated cocktails are $P_1 = \{$TAL1, RPA3, ZFP90, RNF138, LMO2, KLF2$\}$ and $P_2 = \{$DDIT3, MZF1, HOXB3, MIER1, SIRT6, ZNF85$\}$. In the deployed model this holds 222 answer-key combinations out of training while keeping them for evaluation. Single TFs are split with combo-involved TFs forced into training, so a single-TF holdout never silently guts combination coverage. Fate targets are split by fate name into folds to prevent cross-source leakage.

B.4 The two deployed checkpoints

Two checkpoints back the paper, both trained for 60 epochs.

Deployed production world model. Trained with both the single-TF and combination training fractions set to 1.0, so every single TF and every non-answer-key combination is in training and only the 222 answer-key combinations are held out. Like the held-out sibling, it warm-starts from a singles-only base model and oversamples single-TF sentences $8\times$ during the combination finetune, so single-TF effects are retained. The training log records 963,974 cells, a per-epoch reconstruction subsample of 150,000 cells, 495 single-TF blocks and 1622 combination blocks (2117 blocks total), and 22,328 cell sentences per epoch. This is the checkpoint used for all causal-structure validation (App. A numbers and Sec. 3) and as the frozen reward oracle for the planner. Because its training fractions are 1.0, a held-out-singles score on this checkpoint measures memorization and is not a genuine held-out number. Genuine held-out single and combination prediction uses the next checkpoint.

Held-out combination finetune. Trained with a single-TF training fraction of 0.85 and a combination training fraction of 0.8 under a fixed split seed, warm-started from a singles-only base model, with recency-weighted replay of combinations during the finetune. This split yields 432 train and 63 held-out Joung singles, 36 train and 8 held-out Joung combinations, and Briggs combinations split into train and held-out plus the 222 answer-key combinations. Two additional seed variants provide the three seeds behind the main-text prediction p-values. The main-text held-out advantages of $+0.184$ (answer-key combinations) and $+0.225$ (held-out combinations) in drift-removed $\mathrm{Pearson}_{\mathrm{DEG}}$are computed on this checkpoint across the three seeds. The epistasis panels (Fig. 3A–D) also use this held-out sibling for the combination-trained model and its singles-only base for the singles-only arm, so the interaction claims are evaluated on the same recipe as the planner world model. Only the enrichment panel (Fig. 3E) uses the deployed production world model.

B.5 Fate-target annotation

Target fates are defined by a marker-signature pipeline over the Cao fetal cell atlas. We start from the Cao cell-type mean log-expression profiles, restrict to genes on our shared axis, and center them. Cell types are merged into fate groups by average-linkage hierarchical clustering at cosine similarity 0.78 on the centered profiles, and each fate group is named by its dominant tissue–type suffix. For each fate group we take the top 50 genes by one-vs-rest over-expression as its marker set and form a normalized marker signature.

Every candidate perturbation (Briggs combinations, Joung single-TF overexpressions, and Joung arrayed combinations) is summarized by its drift-removed mean response, the residual centroid Δ obtained by subtracting the source-level mean Δ. Each perturbation's residual Δ is scored against all marker signatures and assigned to its argmax fate. A fate is *kept* only if it is *target-separable*, meaning its marker signature is isolated from every other signature (max cosine similarity below 0.75), and *reachable*, meaning it collects at least a minimum number of marker-argmax member

cells (5 for Briggs, 3 for Joung). Each kept fate's target is the drift-removed mean Δ (residual centroid) of its marker-argmax member cells. Kept fates are then tiered. Tier A fates additionally have a positive per-fate cosine silhouette in the marker-gene (specific-gene) subspace, and Tier B fates have a non-positive silhouette. Crucially, the annotation does *not* gate on silhouette. Both tiers are retained, and the silhouette only labels the clean-answer-key core (Tier A) versus the remainder (Tier B). Reachable-fate confidence uses a proper Benjamini–Hochberg correction, which moves the fate set by only one or two Briggs fates relative to a raw permutation cutoff.

Each fate additionally stores its Cao marker genes, its member count, its driver TFs, and its answer-key combinations. The driver and answer-key lists derive from the Briggs P_1/P_2 canonical cocktails and the arrayed-combo table (Sec. B). The pipeline yields 84 fates, namely 65 Briggs, 16 Joung single-TF, and 3 Joung combination fates (Table 2). The zero-shot HSPC fate, the Briggs liver hematopoietic-stem-cell fate (Tier B), is forced to fold -1 and is always held out. Its evaluation uses the dedicated Briggs $P_1 \cup P_2$ answer key rather than Cao markers.

Fates are split into a five-fold leave-fates-out cross-validation partition by *fate name*, so all sources of one fate name share a fold and there is no cross-source leakage. The folds are stratified by name-level tier and assigned round-robin by descending member count, so each Tier A fate is held out exactly once. In each cross-validation fold $f \in \{0, \ldots, 4\}$ the fates assigned to fold f are held out for evaluation and the rest are used for training, and the single fate at fold -1 (HSPC) is held out in every fold. The per-fold held-out counts are 19, 18, 14, 16, 16 for folds 0–4 plus the 1 always-held-out HSPC fate at fold -1.

Table 2: Fate-target counts by source and tier. Tier A fates additionally clear a positive marker-subspace silhouette (clean answer-key core). Tier B fates are target-separable and reachable but silhouette-negative. The HSPC fate (the Briggs liver hematopoietic-stem-cell fate) is one of the 65 Briggs fates, is Tier B, and is forced to fold -1 (always held out).

Source	Fates	Tier A	Tier B
Briggs (combinations)	65	10	55
Joung single-TF	16	2	14
Joung combination	3	2	1
Total	84	14	70

C Training and planner hyperparameters

C.1 World-model hyperparameters

Table 3 lists the exact world-model settings for the deployed model. Training runs in three phases inside the 60 epochs, recorded in the training log. Phase A is epochs 1 to 10, reconstruction only with the directed graph frozen. Phase B is epochs 11 to 50, directed-graph learning active with the full loss. Phase C is epochs 51 to 60, full loss with temperature annealing. The world model trains on CPU high-memory nodes with about 340 GB of RAM.

The architecture is the model of App. A: a per-cell encoder, a latent SEM over d=357 GO:BP program latents solved in closed form, an ESM2 per-TF shift encoder (100-dim descriptor $\rightarrow$ 1024 hidden $\rightarrow$ 357), the shift-mutilate do-operator for k-TF composition, and the masked-GO (SENA) linear decoder to the 4366-gene axis. These same settings train both deployed checkpoints described in Sec. B ("The two deployed checkpoints"). The deployed production world model is the epoch-60 checkpoint, used for all causal-structure validation and as the frozen planner reward oracle. The held-out combination finetune shares every model hyperparameter in Table 3 and differs only in its train fractions (single-TF fraction 0.85, combination fraction 0.8) and its singles-only-base warm-start. It backs the held-out prediction and epistasis panels. Both checkpoints warm-start from a singles-only base and oversample single-TF sentences 8$\times$ during the combination finetune.

C.2 The GFlowNet planner

The deployed planner is a trajectory-balance generative flow network (GFlowNet) over the frozen world model. A cocktail is built one TF at a time by an autoregressive forward policy. Because the

Table 3: World-model hyperparameters for the deployed model. Values not present in the configuration are marked not specified.

Optimizer	Adam	Gene axis G	4366
Learning rate	10^{-3}	Latent programs d	357 (GO:BP mask)
LR warmup epochs	5	Decoder	masked-GO (SENA), λ=0.3
Epochs	60	Shift descriptor	ESM2, 100-dim
Batch size	256	Shift MLP	$100 \to 1024 \to 357$, LeakyReLU(0.2)
Grad accumulation	4	Interaction rank	32
KL β	$0.01 \to 0.5$ over 30 ep	Graph samples (train/eval)	1 / 32
λ_{pert}	10	Mutilation $u_{\max}$, β	0.5, 1.0
DEG-MSE weight, thr	8.0, 0.1	Spectral weight	0.1
Contrastive temp	0.2	Prior / sparse / ortho / acyclicity	0 (prior-off)
Cell-sentence S	64	Sentences per block	4
Sentence batch	32	Single-sentence mult	8
Recon subsample	150,000/epoch	τ anneal	$1.0 \to 0.25$

construction is autoregressive there is a single parent per state, so the backward policy is trivially $P_B \equiv 1$ and the trajectory-balance loss for a terminal cocktail x under goal g is

$$\mathcal{L}_{\text{TB}} = \Big(\log Z(g) + \textstyle\sum_t \log P_F(a_t \mid s_t, g) - \beta \cdot \Phi(x, g) \Big)^2, \tag{7}$$

with reward $R(x) = \exp(\beta\, \Phi(x, g))$ [Bengio et al., 2021] and reward exponent β=48. The deployed flagship uses a *goal-conditioned* partition function $\log Z(g)$, where a per-goal head absorbs the goal-dependent normalizer so the forward policy is free to spread mass across the many high-reward cocktails for each goal. This is the single most important lever for diversity. The alternative of a single shared scalar $\log Z$ for all goals forces the forward policy to carry the goal-dependent normalizer, and it collapses to an amortized argmax at convergence (effective number of distinct sampled sets $\text{ess}_q \approx 1$–2, versus up to the target ceiling of ≈ 46 under the conditional head). We report the shared-$\log Z$ variant as an ablation.

Ensemble-pessimistic reward. The potential $\Phi(x, g)$ is the drift-removed match of the world-model-predicted state to the target fate. To guard against exploiting any single world model's errors, the reward is *ensemble-pessimistic*. We evaluate the same cocktail under K=6 independently trained world models and combine their potentials as

$$\Phi_{\text{pess}}(x, g) = \operatorname*{mean}_{k=1}^{K} \Phi^{(k)}(x, g) \;-\; \lambda \cdot \operatorname*{std}_{k=1}^{K} \Phi^{(k)}(x, g), \qquad \lambda = 0.25, \tag{8}$$

so cocktails on which the ensemble disagrees are penalized. The pessimism weight λ trades recovery against sampling diversity. The deployed value is λ=0.25.

Off-policy replay mixes a fraction γ=0.5 of each batch from a recency-weighted buffer of past cocktails (capacity 5000), which is valid for trajectory balance. The reward oracle is the frozen world-model ensemble, so no surrogate or acquisition proxy is needed. Batches are 16 trajectories per update, trained for 10^6 iterations. The per-source action budget is set at reset, so Joung caps at $k \leq 2$ and Briggs at $k \leq 8$. Prioritized, reward-weighted replay is a reported negative (App. E). The deployed setting is plain recency-weighted replay.

The policy is a set-based autoregressive policy over the vocabulary $\{\text{STOP}\} \cup$ candidates. A shared encoder feeds all slots, so cost grows linearly in k not combinatorially. STOP is forbidden at slot 0 and already-chosen genes are masked at later slots. The action head is a plain multilayer perceptron (MLP) over the per-slot features.

C.3 Design selection

The deployed flagship, the goal-conditioned GFlowNet with the plain MLP head and λ=0.25, was selected by a sweep over the partition function (shared versus goal-conditioned $\log Z$) and the ensemble-pessimism weight $\lambda \in \{0.1, 0.25, 0.5, 0.75, 1.0\}$, at 10^6 iterations across five folds and three seeds on the frozen production world model with a curated candidate pool and per-source k caps. The goal-conditioned $\log Z$ is what enables diverse sampling and λ=0.25 gives the best recovery–diversity trade-off (App. E). The InfoNCE fate-specificity term uses scale β_{contrast}=0.3; a no-contrast control that drops it is reported in App. E.

D Evaluation metrics

This section defines every metric precisely. For a planner the per-fate output is a frequency dictionary of proposed TFs over the best-of-N samples. Picks are ranked by descending selection frequency.

Driver recall at k. With driver set (answer key) $\mathcal{D}$ and the top-k ranked picks $\mathcal{T}_k$, $\mathrm{drv_recall@}k = |\mathcal{T}_k \cap \mathcal{D}|/|\mathcal{D}|$. The denominator is the answer-key size, so this is recall not precision. We report $k \in \{5, 10, 20\}$.

Precision at k. $\mathrm{prec@}k = |\mathcal{T}_k \cap \mathcal{D}|/\min(k, |\mathcal{T}_k|)$. We report $k \in \{5, 10\}$.

Cross-fate discrimination, own minus other. For a scorer s, discrimination is the score of a fate's picks against its own answer key minus the mean score of the same picks against every other held-out fate's answer key, $\mathrm{disc} = s(\text{own}) - \overline{s(\text{other})}$. We report $\mathrm{disc_dr20}$ using recall@20 and $\mathrm{disc_soft1}$ using SOFT-1. The oracle plays the true drivers as picks. For $\mathrm{disc_dr20}$ the oracle equals $1 -$ cross-fate driver-set overlap, so a high oracle means the answer-key driver sets are nearly disjoint across fates and a perfect selector separates them, while a low oracle means fates share drivers and cap what any planner can achieve. A random planner sits near zero because its picks hit own and other drivers equally.

SOFT-1 and SOFT-2. These score in a functional space built from drift-removed, unit-normed single-TF Δ vectors with k-means module labels. SOFT-1 is the frequency-weighted mean, over picks, of the maximum cosine of a pick's single-TF Δ to any driver's single-TF Δ. SOFT-2 is the frequency-weighted fraction of picks whose k-means module coincides with any driver's module. The variants m12, m24, m48 use 12, 24, or 48 modules. These metrics saturate. The nearest-cosine over a broad answer set in a low-dimensional unit-normed space means even random TFs land close to some driver and share one of only twelve modules, so a matched-random null already scores about 0.28 to 0.36 and the policy barely separates. For this reason SOFT-1 and SOFT-2 are reported in the appendix with their null values shown, and the headline uses driver recall, precision, and the own-minus-other discrimination forms.

Best residual match. For each sampled cocktail the world model applies the do-operator, decodes, subtracts the control reconstruction and the drift, and matches the drift-removed predicted Δ against the target on the active dimensions. The best residual match is the maximum match over the best-of-N sampled cocktails.

The matched-random null. For each held-out fate the null records the policy's per-sample cocktail sizes, then draws for each sample a random cocktail of the same size from the candidate pool and recomputes every metric. The reported null value is the mean over R permutations (R=500 for the reported runs). When a policy is gated to a per-goal shortlist, the reported evaluation draws the null from the full 280-candidate pool, which gives a pipeline-versus-ungated contrast consistent across all sources.

The 1000-shuffle-A permutation test. For the causal-structure claim we ablate A inside the frozen do-operator. The variants are original, reversed ($A^\top$), symmetrized ($\frac{1}{2}(A+A^\top)$), and shuffled (a permutation of the off-diagonal entries, which preserves the value multiset and edge count). Every non-original variant is rescaled to match $\|A\|_2$, so magnitude is not a confound. The null draws N=1000 shuffled A matrices, each with a fresh seed, and stores the mean drift-removed $\mathrm{Pearson}_{\mathrm{DEG}}$across perturbations. The single-TF prediction metric is the per-perturbation Pearson of the drift-removed predicted residual against the drift-removed observed residual on the set of DE genes above the threshold 0.05, with the perturbed TF itself excluded so the metric measures downstream recovery. The permutation p-value is the add-one estimate $(1 + \#\{\text{null} \geq \text{original}\})/(N + 1)$, which gives $1/1001 \approx 9.99 \times 10^{-4}$ when no shuffle exceeds the original.

Paired Wilcoxon across fates and Benjamini–Hochberg. The primary planner significance test is a one-sided paired Wilcoxon signed-rank of the policy against its matched-random null across held-out fates. The replicate is the fate when at least two held-out fates exist, with per-fate values averaged over seeds. For a single-fate source such as HSPC the test falls back to pairing across the fold-by-seed training replicates, which gives up to 15 paired points for five folds and three seeds. Benjamini–Hochberg correction is applied across the whole source-by-metric grid. The same one-sided paired Wilcoxon signed-rank test (dropping zero differences) is used for the 66-TF and 14-combination causal-structure tests.

Standard error of the mean (with one delta-degree-of-freedom). Standard errors of the mean, abbreviated s.e.m. in the tables and distinct from the structural equation model above, use the per-seed means as the replicate axis. Within each seed, per-fate values are pooled across folds and averaged, and the standard error of the mean is the sample standard deviation of those per-seed means with one delta-degree-of-freedom divided by $\sqrt{n_{\text{seeds}}}$. A single replicate is reported as undefined rather than zero.

Baselines. MPC-exact is a model-predictive control (MPC) world-model beam search (beam width 8) that forward-selects the cocktail maximizing the predicted drift-removed match. Additive-greedy is a forward selection that sums observed drift-removed single-TF Δ vectors with no world-model interaction. Random-shortlist is a single random pick from the goal shortlist. Every baseline is restricted to the same per-goal shortlist as the policy, so the comparison isolates selection within the pool.

E Full cross-arm planner evaluation

The deployed planner is the goal-conditioned GFlowNet with the plain MLP head and the ensemble-pessimistic reward at λ=0.25 (Eq. (8)), trained over three folds and five seeds. We evaluate it on two axes against non-learned baselines, asking (1) whether it *recovers* the true driver TFs of a held-out fate, and (2) whether it *samples diverse, reward-proportional* cocktails, with a zero-shot HSPC case study. The genuine wins are that the planner is amortized and cheap, is diverse-at-quality, and on the zero-shot HSPC fate covers more of the arrayed answer key with biologically coherent regulators than search does. On per-driver recovery it reaches *parity* with model-predictive beam search, not a win.

Why a GFlowNet trainer. Against a one-step set-PPO trainer, the GFlowNet samples many distinct high-quality cocktails while set-PPO collapses toward a single mode, which is why the deployed planner is a GFlowNet rather than a policy-gradient trainer.

E.1 Axis 1, driver recovery

On the finalized flagship we report held-out Briggs recovery under three readouts of the 256 sampled cocktails, each against its *matched* null (Table 4). The *raw* readout ranks TFs by sampling frequency and is compared to a matched-breadth permutation null (≈ 0.072) that draws random size-matched cocktails and ranks them the same way. The *elite*-10 readout keeps the top 10% of the 256 by reward and ranks within that set, and the *leave-one-out* (LOO) readout ranks each TF by its marginal reward contribution. Both are compared to the *random-plus-elite* null ($\approx$ 0.42), which passes random cocktails through the same reward filter and so measures what the reward filter alone buys. Using the weak permutation null on an elite readout would over-credit the policy for the filter's work, so each readout is matched to its own null. Model-predictive beam search is likewise a reward-guided selector, so the fair comparison between the policy and beam search uses the common random-plus-elite null.

Table 4: Deployed flagship (goal-conditioned GFlowNet, MLP head, ensemble pessimism λ=0.25), held-out Briggs driver recall at 20, three folds $\times$ five seeds. Each readout is scored against its matched null. Here Δ is the policy minus that null. World-model (WM) evaluations per fate index inference cost. The random-plus-elite null is the common reward-guided baseline used to compare the policy against beam search.

Planner / readout	null	abs r@20	Δ vs null	WM evals/fate
policy, raw	permutation (0.072)	0.455	+0.382	256
policy, elite-10	random+elite (0.38)	0.582	+0.200	256
policy, LOO	random+elite (0.42)	0.712	+0.291	207
MPC-exact (beam 8)	random+elite (0.39)	0.507	+0.120	9240
additive	permutation (0.072)	0.012	−0.060	—
random	—	0.071	—	—

The policy adds genuine driver signal above the reward filter. On the elite readout it reaches 0.582, a +0.20 gain over the random-plus-elite null (paired Wilcoxon p=1.7 $\times$ 10^{-5}, n=41 held-out fates),

and on the LOO readout it reaches 0.712 ($+0.29$ above the same null). Beam search reaches 0.507 ($+0.12$ above the filter, p=0.02), so the policy *matches* beam search on absolute recovery. The direct paired comparison of the policy against beam search is non-significant (parity), consistent with the matched, paired analysis. The advantage is compute. The policy uses ≈ 256 world-model evaluations per fate (the LOO readout ≈ 207), while beam search uses ≈ 9240, so the policy attains the same recovery at roughly 36–45$\times$ lower inference cost. Both learned and search methods sit far above the additive (0.012) and random (0.071) baselines, which have no world model.

Table 5 extends the comparison across four metrics on the same elite readout and the same 41 held-out fates. Beam search (a reward-guided selector read out on its own proposals) and the policy are at parity on the broad metrics, recall@20 and soft-recall@20 (paired Wilcoxon p=0.10 and 0.20, not significant), while the policy is significantly ahead at the top of the ranking, recall@5 and precision@5 (both $p < 0.05$); every metric is far above the additive and random baselines ($p < 10^{-4}$). Soft-recall credits functional near-misses through the drift-removed single-TF program space and is scored identically for every method.

Table 5: Per-metric budget-matched comparison on the elite (top-10%-by-reward) readout, n=41 held-out Briggs fates. Baselines are read out on their own reward-selected proposals and restricted to the identical per-fate shortlist; p is a paired per-fate Wilcoxon of the policy against each baseline.

metric	policy	MPC (beam)	additive	random
recall@5	0.357	0.235 (p=0.02)	0.000 ($p<10^{-4}$)	0.049 ($p<10^{-4}$)
recall@20	0.578	0.507 (ns, p=0.10)	0.012 ($p<10^{-4}$)	0.071 ($p<10^{-4}$)
soft-recall@20	0.799	0.716 (ns, p=0.20)	0.089 ($p<10^{-4}$)	0.540 ($p=2\times10^{-4}$)
precision@5	0.198	0.130 (p=0.01)	0.000 ($p<10^{-4}$)	0.020 ($p<10^{-4}$)

E.2 Held-out fate breakdown

The Briggs fate universe is 65 cell-type fates partitioned by a leakage-clean by-name five-fold split (folds 0–4, $\approx$ 13 fates each) with a difficulty tier (A = positive-silhouette separable target, B = separable and reachable). The flagship holds out folds 0, 1, and 2, giving the n=41 held-out fates in Table 6. Fold 0 was evaluated at a single seed and folds 1–2 at five seeds, so fold-0 per-fate values are noisier. The table lists each fate's fold, tier, driver count, and elite recall@20 for the policy (pol) and beam search (MPC), showing the per-fate spread behind the paired test: the policy wins on many fates, ties on some, and loses on others, which is why the aggregate is a statistical tie with beam search.

E.3 Axis 2, diversity at quality

The second axis asks whether the planner returns *many* high-reward cocktails rather than one. On the enumerable HSPC candidate pool we draw 256 proposals from each method and count how many are distinct and above a fixed quality bar (Table 7). Asked to return 256 candidates, budget-matched beam search collapses to a single distinct high-quality cocktail. Size-matched random sampling gives 64 distinct sets but of low reward. The policy returns 154 distinct high-quality cocktails. The policy is the only method that is both diverse and high-quality. This diversity is owned by the goal-conditioned $\log Z$. It lets the effective number of distinct sampled sets approach the target ceiling (ess_q up to ≈ 46), whereas the shared-$\log Z$ variant collapses to an amortized argmax ($\mathrm{ess}_q \approx$ 1–2).

E.4 Zero-shot HSPC, coverage advantage and honest limits

We test the frozen policy zero-shot on a held-out hESC-to-HSPC fate. This is a single fate, so we report it as a case study over the 11 fold$\times$seed runs with no per-fate significance test.

Diversity-coverage advantage on the arrayed key. Against the Briggs arrayed answer key (54 driver TFs across 28 arrayed-validated combinations from the supplementary screen), the policy's diverse elite proposals cover far more of the key than the search or heuristic baselines, an elite recall@50 of 0.131 against ≤ 0.056 for MPC and additive and 0.044 for random, and 15 of the 54 drivers surfaced across the runs against 3–4 for every baseline (Table 8). The surfaced TFs are biologically coherent: the policy proposes the canonical hematopoietic regulators LYL1, KLF1,

Table 6: Per-fate elite recall@20 on the n=41 held-out Briggs fates (folds 0–2).

fold	tier	fate	drv	pol	MPC
-1	B	Liver-Hematopoietic stem cells	4	0.25	0.50
0	B	Heart-Epicardial fat cells	3	1.00	1.00
0	A	Spleen-Mesothelial cells	3	1.00	1.00
0	B	Kidney-Ureteric bud cells	4	0.75	0.75
0	B	Cerebellum-Granule neurons	3	0.67	0.67
0	B	Ciliated epithelial cells	3	0.67	0.67
0	A	Eye-PDE11A_FAM19A2 positive cells	3	0.67	0.33
0	A	Placenta-Trophoblast giant cells	3	0.67	0.67
0	B	Skeletal muscle cells	3	0.67	0.67
0	B	Adrenal-Sympathoblasts	5	0.60	0.20
0	B	Placenta-Syncytiotrophoblasts and villous cytotrophoblasts	2	0.50	0.00
0	B	Placenta-Vascular endothelial cells	3	0.33	1.00
0	B	Vascular endothelial cells	3	0.33	0.67
0	B	Eye-Retinal progenitors and Muller glia	4	0.25	0.25
0	B	Intestine-Stromal cells	2	0.00	0.00
1	B	Pancreas-CCL19_CCL21 positive cells	3	1.00	0.67
1	B	Placenta-Stromal cells	2	1.00	0.50
1	A	Stomach-Squamous epithelial cells	3	0.93	0.67
1	B	Purkinje neurons	2	0.80	0.50
1	B	Stomach-Parietal and chief cells	2	0.80	0.00
1	B	Schwann cells	3	0.73	1.00
1	B	Intestine-Smooth muscle cells	4	0.70	0.50
1	B	Eye-Bipolar cells	3	0.67	0.87
1	B	Kidney-Mesangial cells	3	0.60	0.00
1	B	Pancreas-Ductal cells	3	0.60	0.00
1	B	AFP_ALB positive cells	3	0.53	0.33
1	B	Adrenal-Chromaffin cells	3	0.33	0.33
1	B	Eye-Amacrine cells	2	0.30	0.00
2	A	Heart-Visceral neurons	1	1.00	1.00
2	B	Placenta-IGFBP1_DKK1 positive cells	4	0.90	1.00
2	B	Lung-Bronchiolar and alveolar epithelial cells	3	0.80	0.33
2	B	Intestine-Intestinal epithelial cells	2	0.50	0.50
2	A	Lung-Vascular endothelial cells	2	0.50	0.70
2	B	Stomach-Goblet cells	2	0.50	0.50
2	B	Pancreas-Islet endocrine cells	3	0.47	0.33
2	B	Eye-Astrocytes	2	0.40	0.50
2	B	Liver-Vascular endothelial cells	2	0.40	0.50
2	A	Chromaffin cells	4	0.35	0.75
2	B	Spleen-Vascular endothelial cells	3	0.33	0.67
2	B	Cerebellum-Inhibitory interneurons	4	0.20	0.25
2	B	Adrenal-Vascular endothelial cells	1	0.00	0.00

Table 7: Diversity at quality on the enumerable HSPC pool, 256 proposals per method. Distinct high-quality count is the number of distinct cocktails above a fixed reward bar. Beam search is asked for 256 candidates and returns its frontier. It is not a sampler.

Method	distinct high-quality cocktails	quality
policy (flagship)	154	high
beam-top-256	1	high
random (size-matched)	64	low

KLF2, and IKZF1 in 8–11 of the 11 runs, and members of the blood stem-cell heptad (TAL1/SCL, GATA2, RUNX1, FLI1) and the self-renewal factor HOXB4 in 1–3 runs, all with the regulatory prior switched off.

Honest limits. The advantage is coverage from diverse best-of-N sampling, not per-driver precision. On the narrow twelve-TF $P_1 \cup P_2$ named key, per-run elite recall@20 is 0.12 for the policy against 0.27 for MPC and 0.15 for its random-plus-elite null, so the policy does not beat search or its own reward-filtered null on that key. We do not calibrate a breadth-matched null for the arrayed key, so the arrayed advantage is established against the search baselines but not against equal-breadth random sampling. We therefore treat HSPC as a diversity-coverage result on the broad key, not a beyond-chance per-driver recovery, and note that the reward filter, rather than the policy alone, contributes to surfacing canonical TFs. Zero-shot transfer to a new fate is a data-coverage limit, addressable with in-domain fate data.

E.5 Honest negatives

Two engineering attempts did not help and we report them. Prioritized, reward-weighted replay hurt relative to plain recency-weighted replay, so the deployed GFlowNet uses plain recency-weighted replay. Bulk-signal removal did not lift the cross-fate discrimination wall. The policy's own-minus-other discrimination sits near zero while a drivers-as-picks oracle reaches about 0.65 on Briggs, and the wall persists across reward shaping, exploration budget, and bulk removal, so we report it as a limit of designing against signature-defined fates rather than a bug. The SOFT functional-space

Table 8: Zero-shot HSPC coverage of the 54-TF Briggs arrayed answer key, elite readout, over 11 fold×seed runs (single fate, no p-value). Baselines are read out on their own reward-selected proposals. "surfaced" is the number of distinct arrayed driver TFs appearing across the runs.

Method	recall@50	distinct drivers surfaced (of 54)
policy (flagship)	0.131	15
MPC / greedy	0.056	3
additive	0.056	3
random shortlist	0.044	4

metrics are saturated as described in App. D. On Briggs the policy SOFT-1 is about 0.28 against a null of about 0.28, essentially no separation, which is why SOFT is not a headline.

E.6 Prediction leakage ladder

The world-model prediction advantage over the fair additive baseline survives progressively harder holdouts. We retrained the held-out finetune under two extra rungs, leave-TF-out (rung 2) and leave-program-out (rung 3, leave-P_1-out), and re-measured drift-removed $\mathrm{Pearson}_{\mathrm{DEG}}$. Table 9 shows that on held-out Briggs combinations the world model with its interaction head beats both the observed-additive and the model-additive baselines at every rung, and the margin holds at high order k6–8. The honest counterpoint is the answer-key row. The 222 high-order answer-key combinations are extrapolations the world model does not predict better than observed-additive (0.133 versus 0.208 at rung2), consistent with the main-text limitation that composition is effectively additive for held-out HSPC TFs that never appear in the training combinations.

Table 9: Prediction leakage ladder, drift-removed $\mathrm{Pearson}_{\mathrm{DEG}}$, world model with head versus observed-additive and model-additive baselines. Rung labels are the progressively harder retrain holdouts. Values are group means with n combinations.

Rung / group	n	WM+head	WM no-head	obs-additive	model-additive
Rung2, leave-TF-out retrain					
Briggs combos train (in-dist)	387	0.623	0.385	0.206	0.294
Briggs combos held-out (set-out)	97	0.464	0.358	0.245	0.280
Briggs held-out k6–8	6	0.478	0.087	0.096	0.064
Joung combos held-out	8	0.957	0.397	0.134	0.100
Briggs answer-key (extrapol.)	222	0.133	0.119	0.208	0.111
Rung3, leave-program-out retrain					
Briggs combos train (in-dist)	602	0.591	0.381	0.197	0.259
Briggs combos held-out	151	0.459	0.335	0.224	0.231
Briggs held-out k6–8	26	0.418	0.115	0.122	0.054

F Additional world-model results

F.1 Reconstruction, discrimination, and per-DEG behavior

The deployed production model reaches a stable reconstruction MSE of about 0.52 and a KL of about 0.26 at epoch 60 with the acyclicity residual at zero throughout (training log). Drift-removed discrimination on Joung singles is near ceiling in the finetune checkpoint, with held-out singles reaching AUROC 0.917 at rung2 and 0.952 at rung3 while the observed-additive control is 1.0 by construction. The world model's advantage grows with combination order. On held-out Briggs combinations the model-with-head $\mathrm{Pearson}_{\mathrm{DEG}}$stays high at k6–8 (0.478 at rung2) while both additive baselines collapse toward 0.06 to 0.12 (Table 9), which is the signature of genuine non-additive structure at high order.

F.2 The directed causal graph is load-bearing for prediction

The causal-structure panels of Fig. 3 (E, F) in the main text are the result this section expands. We ask a sharp, testable question about the learned graph A, namely whether its *direction* is load-bearing

for prediction. We answer by ablating A inside the frozen do-operator (reversing, symmetrizing, or shuffling its edges) and re-measuring Δ-prediction on the deployed production world model. This is a relative true-vs-shuffled comparison on the same perturbations, so it needs no held-out split.

Panels A and B show single-TF and combination prediction under original, reversed, symmetrized, and shuffled A, scored by drift-removed $\mathrm{Pearson}_{\mathrm{DEG}}$and cross-fate discrimination AUROC. Against a 1000-shuffle null, single-TF prediction collapses from Pearson 0.536 $[0.443, 0.625]$ to a shuffled mean of -0.156 ± 0.010, a gap of $+0.692$ that no shuffle reaches (permutation $p < 0.001$, paired Wilcoxon across 66 TFs p=8.9×10^{-11}, with 59 of 66 TFs predicting worse under a shuffled graph). Discrimination AUROC falls from 0.84 to 0.60 and top-1 driver recall from 0.73 to 0.02. Reversing the edges is as damaging as shuffling them, so the effect is directional rather than a matter of connectivity alone. On combinations the gap is smaller ($+0.102$, permutation $p < 0.001$, Wilcoxon p=6.1×10^{-5}, all 14 combinations improving) because the drift-removed combo response is dominated by a shared high-order shift. We therefore quote the single-TF effect as the headline. The full ablation numbers are in Table 10.

Panel C shows that the latent graph is biologically structured. The masked decoder pins each latent axis to a named GO:BP program at mask coherence 1.00, and the highest-connectivity hubs of A are developmental programs. We do *not* claim A recovers a curated GRN. With A=0 the do-operator injects only the direct shift S_g, which already ranks a TF's direct targets, so A=0 stays competitive on a direct-target metric. Propagation over the directed A is what carries the genuine downstream and combination response, which is where the directed graph is load-bearing.

F.3 The edge-direction ablation in full

Table 10 gives the complete edge-direction ablation, computed on the deployed production world model. For single TFs the drift-removed $\mathrm{Pearson}_{\mathrm{DEG}}$collapses from 0.536 with 95% bootstrap CI $[0.443, 0.625]$ to a shuffled-A mean of -0.156 ± 0.010 over 1000 shuffles, a gap of $+0.692$ with permutation $p < 0.001$ (no shuffle exceeds the original) and paired Wilcoxon $p = 8.9 \times 10^{-11}$ across 66 TFs. Reversing the edges (-0.130) is about as damaging as shuffling them, which shows the effect is directional and not merely connectivity. Top-1 driver recall collapses from 0.727 to 0.015 and discrimination AUROC from 0.844 to 0.600. On combinations the raw Pearson is drift-dominated (original 0.977), so the gap is smaller at $+0.102$ but still significant (permutation $p < 0.001$, Wilcoxon $p = 6.1 \times 10^{-5}$ across 14 combinations, all improve). We therefore quote the single-TF gap as the headline and do not carry $+0.69$ over to combinations.

Table 10: Edge-direction ablation on the deployed model, drift-removed $\mathrm{Pearson}_{\mathrm{DEG}}$, top-1 driver recall, and discrimination AUROC under original, reversed, symmetrized, and shuffled A. Non-original variants are rescaled to match $\|A\|_2$.

	original	reversed	symmetrized	shuffled
Single Pearson (n=66)	0.536	−0.130	0.107	−0.156
Single recall@1	0.727	0.015	0.076	0.015
Single disc AUROC	0.844	0.592	0.686	0.600
Combo Pearson (n=14)	0.977	0.797	0.918	0.875
Combo recall@1	0.429	0.071	0.071	0.057

F.4 The retired "$+0.68$" result

An earlier combination-prediction figure of about $+0.68$ over additive came from a non-deployable checkpoint and a raw, non-drift-removed Pearson against a cross-context additive baseline that itself scored near zero. That comparison conflated cross-context drift with model skill, because a large part of the raw combination Pearson is the shared mean shift that any model captures. Once we remove drift and compare against a fair, same-context additive baseline, the honest advantage is the $+0.18$ to $+0.23$ reported in the main text and corroborated by the leakage ladder in Table 9. We retire the $+0.68$ figure.

F.5 Magnitude over-prediction and calibration

The interaction sign and ranking are recovered but the interaction magnitude is uncalibrated. The DEG-MSE term is deliberately unwhitened so the model does track per-gene response magnitude, yet on epistasis the recovered interaction carries a sub-additive offset rather than a clean scale, so we report sign and rank recovery and flag the magnitude as an honest calibration limit. This matches the leave-one-out cross-validated pairwise interaction correlation of $+0.39 \pm 0.088$ reported in the main text, which measures rank and sign agreement rather than calibrated magnitude.

F.6 Identifiability and rotational ambiguity

Fixing the decoder to GO programs pins each latent axis to a named biological process and removes the coordinate permutation ambiguity of the latent, following the identifiable causal-disentanglement lineage of Discrepancy-VAE and its GO-structured variant [Zhang et al., 2023, de la Fuente et al., 2025]. We do not claim full structural identifiability of A. The effective decoder that maps the exogenous term to genes is $D_{\text{eff}} = D\big(I - \operatorname{diag}(1-u)A^{\top}\big)^{-1}$, and any invertible transform T with $A'^{\top} = TA^{\top}T^{-1}$ and $D' = DT^{-1}$ that preserves the masked structure leaves the observational and interventional distributions unchanged. This is a rotational ambiguity in A, which we treat as a known caveat. Our load-bearing claim is the weaker and testable one. The graph direction improves prediction under the edge-direction ablation (Table 10), and the do-operator is exact for composition (Eq. 2). We do not claim that A is a recovered gene-regulatory network. Because the model is trained prior-off on only TFs, TF-to-target databases are off-predictors in our regime, and a plain co-expression baseline edges out the projected graph on every reference, as expected for a context-specific prior-off model [Nourisa et al., 2025, Müller-Dott et al., 2023, Garcia-Alonso et al., 2019, Szklarczyk et al., 2023].

G Fate-space annotation QC

Appendix B defines the fate targets and their counts. Here we give the keep-rule that decides which merged groups become fates. We keep a merged group only if it is a genuine and separable mode, which we test by a positive mean per-fate silhouette in the drift-removed program space together with a minimum-member count. We iterate this rule to convergence and report robustness over a sweep of the merge cut and the minimum-member count. This data-driven rule replaces an arbitrary isolation cutoff and applies uniformly across the sources.

H Causal validation: significance and per-TF breakdown

The main-text edge-direction ablation (Fig. 3E,F) reports a point gap. Here we give its null distribution and its per-TF consistency. Fig. 6A places the observed single-TF mean Δ-Pearson ($+0.54$, 95% CI shaded) against a 1000-shuffle-A null, which no shuffle reaches ($0/1000$ exceed, permutation $p < 0.001$, paired Wilcoxon across 66 TFs $p{=}8.9 \times 10^{-11}$). Fig. 6B breaks the ablation out per held-out TF, where 59 of 66 TFs individually predict worse under a shuffled graph, so the effect is not an averaging artifact.

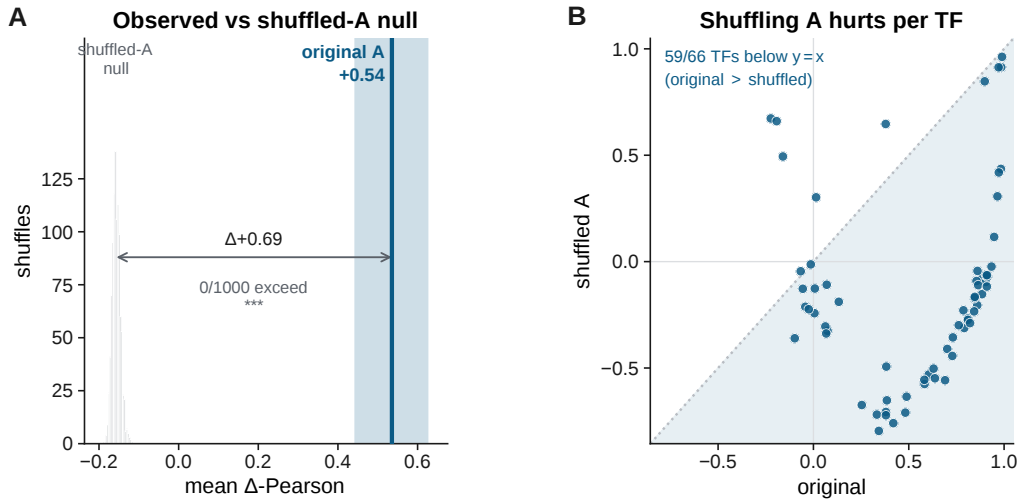


Figure 6: **Edge-direction ablation, significance and per-TF breakdown.** (A) 1000-shuffle-A null against the observed single-TF mean Δ-Pearson ($0/1000$ exceed, $p < 0.001$). (B) Per-TF original against shuffled Δ-Pearson, with 59 of 66 below the diagonal.

I Planner training and generalization

Fig. 7A shows the goal-conditioned GFlowNet training dynamics. The trajectory-balance objective converges and the policy keeps drawing from nearly the full candidate pool (about 280 distinct TFs), so it does not mode-collapse at the TF level, and the cocktail-level diversity is quantified in App. E. Fig. 7B compares in-domain against held-out driver recall by source. Held-out performance tracks in-domain, so the planner does not overfit the training fates.

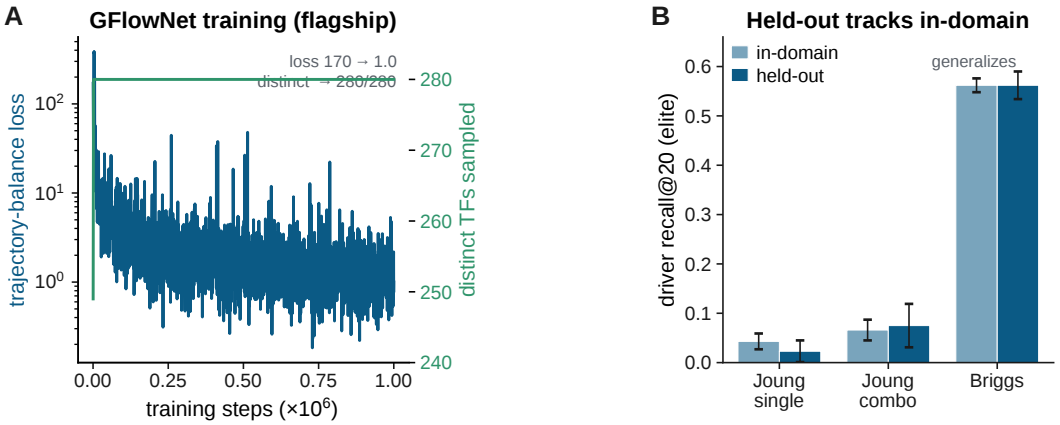


Figure 7: **Planner training and generalization.** (A) Trajectory-balance loss and distinct-TF count over training for the flagship conditional GFlowNet (loss $170 \to 1$, distinct TFs $\to 280/280$). (B) In-domain against held-out driver recall@20 (elite readout) per source. Briggs generalizes cleanly (0.56 held-out), while the two small Joung sources (six and two held-out fates) stay near the floor.

J Broader impacts

This work provides a computational method for proposing transcription-factor cocktails that steer cell fate. Its intended positive impact is to speed up hypothesis generation for cellular reprogramming and regenerative medicine and to reduce the combinatorial screening burden. The potential

negative impacts are limited. The outputs are in-silico hypotheses derived from public single-cell datasets, and they require wet-lab validation before any use, so there is no direct path to harmful deployment. The main practical risk is misallocated experimental effort if the predictions are over-trusted. We mitigate this risk by reporting calibrated error bars and significance, by stating honest negatives, and by delimiting where the method fails (Sec. 7).

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